

Urban Heat Stress Risk Forecasting System for Pakistan: A Machine Learning Approach

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Abstract

Climate change has intensified urban heat stress, particularly affecting rapidly urbanizing regions like Pakistan. This project presents a comprehensive machine learning-based system for forecasting monthly heat stress risk across seven major Pakistani cities (Karachi, Lahore, Multan, Islamabad, Rawalpindi, Peshawar, and Quetta) spanning 1974–2023. We integrate four heterogeneous public datasets: daily weather observations, World Bank population density and urbanization metrics, and global surface temperature records. Through systematic feature engineering, we construct a composite Heat Stress Index combining temperature, population, urbanization, and wind factors. We develop two complementary models: a Logistic Regression monitoring model achieving 92.3% accuracy and 95.8% recall on extreme events, and a Histogram Gradient Boosting forecasting model with 92.0% accuracy for climate-scenario projections. The system classifies risk into four categories (Low, Moderate, High, Extreme) and provides probabilistic forecasts up to 72 months ahead under baseline, +1°C, and +2°C warming scenarios. An interactive Streamlit dashboard enables stakeholders to explore city-specific risks, compare scenarios, and understand model predictions through SHAP-based explainability. Comprehensive error analysis reveals that most misclassifications occur at adjacent class boundaries, with the system reliably detecting 96% of extreme heat events. This work demonstrates the viability of classical machine learning for operational heat health surveillance in data-constrained settings, providing actionable intelligence for public health planning and climate adaptation strategies.

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1 Introduction

1.1 Background and Motivation

Urban heat stress represents one of the most pressing public health challenges of the 21st century, with climate change amplifying the frequency, intensity, and duration of extreme heat events worldwide [1]. Pakistan, as a rapidly urbanizing nation with a population exceeding 240 million, faces acute vulnerability to heat-related health impacts. The country has experienced several catastrophic heat waves in recent decades, including the devastating 2015 Karachi heat wave that claimed over 1,200 lives [2], and the unprecedented nationwide heat crisis of 2022–2023 that affected millions across Punjab and Sindh provinces.

The urban heat island effect exacerbates natural temperature extremes in Pakistani cities, where rapid development, increasing population density, and limited green infrastructure create localized warming of 2–5°C above surrounding rural areas [3]. Compounding these challenges, Pakistan’s geographic diversity—spanning coastal plains, river valleys, mountainous regions, and high-altitude plateaus—creates spatially heterogeneous climate patterns requiring city-specific risk assessment approaches.

Traditional heat warning systems rely on absolute temperature thresholds or generic heat indices that fail to account for local climatological adaptation, socioeconomic vulnerability, and temporal dynamics. Furthermore, existing systems typically operate reactively, issuing alerts only when dangerous conditions materialize, leaving insufficient time for preventive interventions. There exists a critical need for proactive, data-driven forecasting systems that can predict heat stress risk weeks to months in advance, enabling public health authorities, urban planners, and policymakers to implement timely protective measures.

1.2 Problem Statement

This project addresses the following research problem:

How can we develop an accurate, interpretable, and operationally viable machine learning system to forecast monthly urban heat stress risk for major Pakistani cities across multiple time horizons (6–72 months) under various climate scenarios, while accounting for local climatology, urbanization patterns, and temporal dependencies?

The problem encompasses several technical challenges:

1. **Data Integration:** Combining heterogeneous data sources with different temporal resolutions, spatial granularities, and completeness levels into a unified modeling framework.
2. **Risk Quantification:** Defining a composite heat stress metric that captures physiological, environmental, and socioeconomic dimensions of vulnerability.
3. **Temporal Modeling:** Building forecasting models that respect temporal ordering, capture seasonal patterns, and generalize to future climate conditions outside historical experience.
4. **Class Imbalance:** Handling the relative rarity of extreme events while maintaining high recall for safety-critical predictions.

5. **Interpretability:** Ensuring model predictions are explainable to domain experts and decision-makers who may lack machine learning expertise.
6. **Scenario Analysis:** Enabling exploration of alternative climate futures (+1°C, +2°C warming) to support long-term planning and policy development.

1.3 Objectives

The primary objectives of this project are:

1. Develop a robust data pipeline integrating weather observations, demographic trends, and climate indicators for seven major Pakistani cities.
2. Engineer a comprehensive Heat Stress Index incorporating temperature, population density, urbanization, and meteorological factors.
3. Train and evaluate multiple machine learning models for both real-time monitoring and forward-looking forecasting scenarios.
4. Achieve high accuracy (>90%) and extreme event recall (>95%) to ensure reliable detection of dangerous heat conditions.
5. Implement an interactive visualization dashboard enabling stakeholders to explore predictions, compare scenarios, and understand model behavior.
6. Conduct thorough error analysis to identify failure modes, quantify uncertainty, and establish trust in operational deployment.
7. Provide actionable recommendations for public health interventions, urban planning, and climate adaptation strategies.

1.4 Scope and Contributions

This work focuses on monthly heat stress risk forecasting for seven major Pakistani cities with sufficient historical data coverage. The system operates at the city level rather than neighborhood or district granularity, using publicly available datasets that can be regularly updated. Key contributions include:

- A reproducible pipeline for processing 50 years of heterogeneous climate and socioeconomic data.
- A validated Heat Stress Index tailored to Pakistani urban contexts.
- Comparative evaluation of classical machine learning approaches (Logistic Regression, Decision Trees, Random Forest, Histogram Gradient Boosting) for heat risk classification.
- Rigorous temporal cross-validation and held-out testing ensuring realistic performance estimates.
- SHAP-based explainability analysis revealing which features drive high-risk predictions.

- An operational dashboard prototype demonstrating the feasibility of integrating ML forecasts into decision support workflows.

The system is designed to be extended with additional cities, finer temporal resolutions, and integration with health outcome data as they become available.

1.5 Organization of Report

The remainder of this report is structured as follows: Section 2 reviews related work on heat stress modeling and forecasting. Section 3 formally defines the prediction problem and evaluation criteria. Section 4 describes data sources and acquisition processes. Section 5 details preprocessing steps and quality assurance. Section 6 presents feature engineering methodologies. Section 7 explains the machine learning pipeline and model architectures. Section 8 documents experimental setup and hyperparameter tuning. Section 9 presents comprehensive results and performance analysis. Section 10 describes the dashboard proof-of-concept. Section 11 discusses findings, limitations, and implications. Section 12 outlines future research directions. Section 13 concludes, and references and appendices follow.

2 Related Work

2.1 Heat Stress Indices and Metrics

Quantifying heat stress has been a subject of extensive research across meteorology, physiology, and public health. The National Weather Service’s Heat Index [4] combines temperature and relative humidity to estimate apparent temperature, widely used in operational warnings. The Wet Bulb Globe Temperature (WBGT) incorporates solar radiation and wind effects, commonly used for occupational safety [5]. More recent indices like the Universal Thermal Climate Index (UTCI) [6] account for human thermoregulation dynamics.

However, these indices focus primarily on physiological responses to immediate meteorological conditions, neglecting socioeconomic vulnerability dimensions. Our Heat Stress Index extends beyond meteorology by incorporating population density and urbanization as proxies for exposure and adaptive capacity, recognizing that heat health impacts depend on both hazard intensity and community resilience.

2.2 Machine Learning for Climate Prediction

Machine learning has been increasingly applied to climate and weather forecasting problems. Classical approaches like Support Vector Machines and Random Forests have shown success in temperature prediction tasks [7]. Deep learning methods, particularly Long Short-Term Memory (LSTM) networks and Convolutional Neural Networks, have achieved state-of-the-art results for time series forecasting [8].

For heat-related health prediction specifically, several studies have employed ML techniques. McGeehin and Mirabelli [9] used logistic regression to model heat-related mortality. More recently, Basu et al. [10] applied ensemble methods to predict hospital admissions during heat waves. However, most existing work focuses on short-term (1–3

day) forecasting or retrospective analysis rather than medium-term (weeks to months) forward-looking risk assessment.

Our approach bridges this gap by developing models suitable for both real-time monitoring and scenario-based forecasting horizons extending to six years, enabling proactive rather than reactive decision-making.

2.3 Urban Heat Island Studies in Pakistan

Research on urban heat in Pakistan has predominantly focused on remote sensing analyses of surface urban heat islands. Waqar et al. [11] analyzed Landsat imagery to quantify land surface temperature increases in Islamabad associated with urban expansion. Similarly, Ullah et al. [12] examined Lahore’s heat island intensity and its correlation with land use patterns.

While these studies provide valuable spatial characterizations, they lack the temporal forecasting component necessary for operational warning systems. Furthermore, they typically do not integrate socioeconomic dimensions or provide health risk classifications. Our work complements remote sensing approaches by developing a time-series forecasting framework using in-situ weather observations combined with demographic trends.

2.4 Heat Warning Systems

Operational heat warning systems exist in many developed countries. The US National Weather Service issues heat advisories based on Heat Index thresholds [13]. France implemented a national heat health watch-warning system after the 2003 heat wave [14], featuring a four-level alert system similar conceptually to our risk categories.

Pakistan currently lacks a comprehensive national heat warning system, with only ad-hoc alerts issued during extreme events. Our system provides a data-driven foundation for establishing systematic heat surveillance and early warning capabilities tailored to Pakistani urban contexts.

3 Problem Definition

3.1 Formal Problem Statement

Let $\mathcal{C} = \{c_1, c_2, \dots, c_7\}$ denote the set of seven major Pakistani cities under analysis. For each city $c \in \mathcal{C}$ and time index t representing a specific month, we aim to predict a discrete risk label $y_t \in \{0, 1, 2, 3\}$ corresponding to risk categories:

$$y_t = \begin{cases} 0 & \text{Low Risk} \\ 1 & \text{Moderate Risk} \\ 2 & \text{High Risk} \\ 3 & \text{Extreme Risk} \end{cases} \quad (1)$$

Each observation is characterized by a feature vector $\mathbf{x}_t \in \mathbb{R}^d$ comprising meteorological variables, temporal indicators, demographic trends, and derived quantities. The learning task is to estimate a function $f : \mathbb{R}^d \rightarrow \{0, 1, 2, 3\}$ that minimizes prediction

error on future time periods while maintaining interpretability and high recall on the extreme class.

Formally, for a training dataset $\mathcal{D}_{\text{train}} = \{(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_n, y_n)\}$ with samples ordered temporally, we seek:

$$f^* = \arg \min_{f \in \mathcal{F}} \mathbb{E}_{(\mathbf{x}, y) \sim \mathcal{D}_{\text{test}}} [\mathcal{L}(f(\mathbf{x}), y)] \quad (2)$$

where $\mathcal{F}$ is the hypothesis space, $\mathcal{L}$ is a loss function (cross-entropy for training, macro-F1 for evaluation), and $\mathcal{D}_{\text{test}}$ consists of strictly future months unseen during training.

3.2 Risk Category Definitions

Risk categories are defined based on percentile thresholds of the Heat Stress Index (HSI), which combines standardized climate and demographic variables:

$$\text{Low} : \text{HSI} < P_{50} \quad (3)$$

$$\text{Moderate} : P_{50} \leq \text{HSI} < P_{75} \quad (4)$$

$$\text{High} : P_{75} \leq \text{HSI} < P_{90} \quad (5)$$

$$\text{Extreme} : \text{HSI} \geq P_{90} \quad (6)$$

where P_{50} , P_{75} , and P_{90} represent the 50th, 75th, and 90th percentiles of the HSI computed over the training period. This relative definition ensures risk categories are calibrated to local climatology while being sensitive to deviations from typical conditions.

3.3 Evaluation Metrics

Given the imbalanced nature of risk categories and the critical importance of detecting extreme events, we employ multiple evaluation metrics:

1. **Accuracy:** Overall correctness across all classes

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (7)$$

2. **Macro-F1 Score:** Harmonic mean of precision and recall, averaged across classes without weighting by frequency

$$F1_{\text{macro}} = \frac{1}{K} \sum_{k=1}^K \frac{2 \cdot \text{Precision}_k \cdot \text{Recall}_k}{\text{Precision}_k + \text{Recall}_k} \quad (8)$$

3. **Extreme Class Recall:** Proportion of true extreme events correctly identified

$$\text{Recall}_{\text{extreme}} = \frac{TP_{\text{extreme}}}{TP_{\text{extreme}} + FN_{\text{extreme}}} \quad (9)$$

4. **Confusion Matrix Analysis:** Detailed examination of misclassification patterns to understand whether errors occur between adjacent or non-adjacent classes.

Macro-F1 serves as the primary optimization objective during model selection, with extreme class recall as a critical constraint (target: $\geq 90\%$) to ensure safety for public health applications.

3.4 Forecasting Scenarios

The system supports three forecasting scenarios to enable climate adaptation planning:

1. **Baseline:** Historical climate patterns projected forward using recent 5-year trends in temperature and demographic variables.
2. **+1°C Warming:** Baseline scenario with uniform 1°C increase in temperature variables, representing moderate climate change.
3. **+2°C Warming:** Baseline scenario with uniform 2°C increase, representing more severe warming consistent with IPCC RCP4.5 scenario [21].

For each scenario, predictions are generated at multiple forecast horizons: 6, 12, 24, 36, 48, 60, and 72 months ahead, enabling both short-term preparedness and long-term strategic planning.

4 Data Acquisition and Preparation

4.1 Data Sources Overview

This project integrates four primary data sources, each providing complementary information about climate conditions, urbanization trends, and demographic dynamics. Table 1 summarizes the key characteristics of each dataset.

Table 1: Summary of data sources used in the forecasting system

Dataset	Temporal Span	Records	Resolution	Source
Pakistan City Weather	1974–2023	157,958	Daily, 7 cities	Meteostat API
Population Density	1960–2023	271	Annual, National	World Bank
Urbanization %	1960–2023	271	Annual, National	World Bank
Surface Temperature	1800–2023	219,603	Monthly, National	NOAA/NASA

4.2 Pakistan City Weather Data

4.2.1 Description and Coverage

The primary climate dataset comprises daily meteorological observations for seven major Pakistani cities obtained from the Meteostat API [15], which aggregates data from official weather stations maintained by the Pakistan Meteorological Department and international partners. The dataset spans 50 years (1974–2023) and includes the following variables:

- Temperature: Average (tavg), minimum (tmin), maximum (tmax) in °C
- Precipitation: Daily total (prcp) in mm, snowfall (snow) in mm
- Wind: Speed (wspd) in km/h, peak gust (wpgt) in km/h, direction (wdir) in degrees
- Atmospheric: Pressure (pres) in hPa

- Sunshine: Duration (tsun) in hours

City selection criteria ensured robust historical coverage:

1. Minimum 600 months (≈ 50 years) of temperature data
2. At least 80% data completeness for average temperature (tavg)
3. Geographic diversity representing Pakistan’s climatic zones

The seven selected cities and their characteristics are shown in Table 2.

Table 2: Geographic and climatic characteristics of study cities

City	Latitude	Longitude	Elevation (m)	Province	Climate Type
Karachi	24.86°N	67.03°E	8	Sindh	Hot desert
Lahore	31.54°N	74.35°E	217	Punjab	Humid subtropical
Multan	30.19°N	71.45°E	122	Punjab	Hot desert
Islamabad	33.73°N	73.17°E	508	Federal	Humid subtropical
Rawalpindi	33.60°N	73.22°E	508	Punjab	Humid subtropical
Peshawar	34.52°N	71.58°E	359	Khyber Pakhtunkhwa	Semi-arid
Quetta	30.19°N	67.00°E	1654	Balochistan	Cold semi-arid

4.2.2 Data Quality Assessment

Initial data quality analysis revealed varying completeness levels across cities and variables (Table 3). Temperature variables (tavg, tmin, tmax) exhibited the highest completeness (85–95%), while wind speed and sunshine duration had substantial missing values (40–60% missing).

Table 3: Data completeness by variable across all cities

Variable	Non-null %	Handling Strategy
tavg (Avg Temperature)	92.3%	Forward-fill + interpolation
tmin (Min Temperature)	88.7%	Forward-fill + interpolation
tmax (Max Temperature)	91.5%	Forward-fill + interpolation
prcp (Precipitation)	78.4%	Fill with 0 (no rain)
wspd (Wind Speed)	56.2%	Median imputation
pres (Pressure)	64.8%	Forward-fill
tsun (Sunshine)	42.1%	Forward-fill

4.3 World Bank Demographic Data

4.3.1 Population Density

Annual population density estimates for Pakistan were obtained from the World Bank Open Data portal [16], indicator code EN.POP.DNST (Population density, people per sq. km of land area). The dataset provides country-level aggregates from 1960–2023, enabling analysis of long-term demographic trends.

Pakistan’s population density increased from approximately 45 people/km² in 1960 to over 300 people/km² in 2023, reflecting rapid population growth and urbanization. For modeling purposes, annual values were linearly interpolated to monthly resolution and merged with weather data by year-month.

4.3.2 Urbanization Percentage

Urban population as a percentage of total population was obtained from World Bank indicator `SP.URB.TOTL.IN.ZS` [16]. This metric captures the pace of urbanization, which correlates with increased impervious surfaces, reduced vegetation, and amplified urban heat island effects.

Pakistan’s urbanization rate grew from approximately 22% in 1960 to 38.5% in 2023, one of the highest urbanization rates in South Asia. Similar to population density, annual values were interpolated to monthly resolution.

4.4 Global Surface Temperature

Monthly average surface temperature data for Pakistan were obtained from the NOAA/-NASA collaborative dataset [17], which provides reconstructed historical temperatures based on station observations and reanalysis products. This dataset spans 1800–2023 and offers a national-level climate signal complementing city-specific weather data.

The global surface temperature variable enables separation of local weather variability from broader climate trends, allowing the model to distinguish between anomalous heat due to weather patterns versus secular warming.

4.5 Data Integration Strategy

Integrating data sources with different temporal resolutions (daily weather, annual demographics, monthly climate) required careful alignment:

1. **Daily to Monthly Aggregation:** Weather variables were aggregated to monthly means (temperature, pressure, wind), sums (precipitation, sunshine), and extremes (minimum `tmin`, maximum `tmax`) per city.
2. **Annual to Monthly Expansion:** Population density and urbanization percentages were repeated for all months within each year, effectively treating them as constant within-year but varying across years.
3. **Temporal Alignment:** All datasets were merged on `(city, year, month)` tuples, with missing months removed to prevent spurious records.
4. **Coverage Filtering:** Only city-year-month combinations with at least 20 days of weather observations were retained to ensure representative monthly statistics.

The final integrated dataset contains 9,168 monthly observations across seven cities (Table 4).

Table 4: Final integrated dataset structure

Dimension	Value
Cities	7
Time Period	1974–2023 (50 years)
Total Months	600
Total City-Month Records	9,168
Features (after engineering)	27
Target Classes	4 (Low, Moderate, High, Extreme)

5 Preprocessing and Quality Assurance

5.1 Daily to Monthly Aggregation

Raw daily weather data were aggregated to monthly resolution to match the decision-making timescale for public health interventions (weeks to months rather than hours to days). The aggregation scheme balances information retention with noise reduction:

Temperature Variables:

$$\text{tavg_mean}_m = \frac{1}{n_m} \sum_{d \in m} \text{tavg}_d \quad (10)$$

$$\text{tmin_min}_m = \min_{d \in m} \text{tmin}_d \quad (11)$$

$$\text{tmax_max}_m = \max_{d \in m} \text{tmax}_d \quad (12)$$

where m denotes a month and n_m is the number of days with valid observations.

Precipitation and Sunshine:

$$\text{prcp_sum}_m = \sum_{d \in m} \text{prcp}_d \quad (13)$$

$$\text{tsun_sum}_m = \sum_{d \in m} \text{tsun}_d \quad (14)$$

Wind and Pressure: Mean values were computed for wind speed and atmospheric pressure to capture typical monthly conditions.

5.2 Missing Data Imputation

Despite high overall completeness, localized gaps existed due to station maintenance, equipment failures, or data transmission issues. A hierarchical imputation strategy was employed:

1. **Forward Fill:** Carry forward the last valid observation for up to 3 consecutive missing months, appropriate for slowly-varying variables (pressure, seasonal temperature patterns).
2. **Linear Interpolation:** For gaps exceeding 3 months but within a continuous series, linearly interpolate between surrounding valid values.

3. **Seasonal Median:** For remaining gaps, substitute with the median value for that calendar month computed over the entire time series (e.g., all Januaries).
4. **Exclusion:** Months with <20 valid days or critical variable(s) entirely missing were excluded from analysis ($\approx 8\%$ of potential records).

This strategy prioritizes preservation of temporal continuity while avoiding imputation of implausible values.

5.3 Outlier Detection and Treatment

Extreme values were scrutinized to distinguish genuine climatic extremes from measurement errors:

- **Temperature:** Values beyond ± 4 standard deviations from city-specific seasonal means were flagged. Manual verification against historical records confirmed genuine extremes (e.g., Jacobabad’s 54°C reading in 2017 [18]).
- **Precipitation:** Daily values exceeding 300 mm were verified against regional flood records. Isolated erroneous readings (e.g., sensor malfunctions reporting 999 mm) were removed.
- **Wind Speed:** Peak gusts exceeding 150 km/h were checked against storm reports. Values attributed to tropical cyclones (particularly affecting Karachi) were retained.

Less than 0.3% of observations were flagged as erroneous and replaced via interpolation.

5.4 City Filtering Criteria

To ensure robust model training, cities were required to meet minimum coverage thresholds:

- ≥ 600 months (50 years) of records
- $\geq 80\%$ completeness for average temperature (tavg)
- $\geq 60\%$ completeness for maximum temperature (tmax)

These criteria excluded smaller cities with insufficient historical data, focusing analysis on major urban centers with reliable long-term observations. All seven cities in Table 2 met these requirements.

5.5 Temporal Consistency Checks

Several validation checks ensured temporal integrity:

1. **Duplicate Detection:** Confirmed no duplicate (city, date) pairs existed after aggregation.
2. **Chronological Ordering:** Verified monotonic increase of timestamps within each city’s time series.

3. **Month Completeness:** Identified and flagged month-years with <20 daily observations to prevent biased monthly statistics.
4. **Seasonal Plausibility:** Checked that temperature seasonality aligned with expected patterns (e.g., Karachi’s July temperatures not exceeding January temperatures by $>20^\circ\text{C}$).

6 Feature Engineering

Feature engineering transforms raw measurements into representations better suited for machine learning. Our approach combines domain knowledge about heat stress mechanisms with data-driven feature discovery.

6.1 Temporal Features

Cyclical temporal patterns were encoded using sinusoidal transformations to avoid discontinuities between December (month 12) and January (month 1):

$$\text{month_sin} = \sin\left(\frac{2\pi \cdot \text{month}}{12}\right) \quad (15)$$

$$\text{month_cos} = \cos\left(\frac{2\pi \cdot \text{month}}{12}\right) \quad (16)$$

This representation allows models to learn that January follows December smoothly rather than treating them as distant categorical values (1 vs. 12). Additionally, a linear `year_scaled` variable captures secular trends, though excluded from the final forecasting model to prevent extrapolation artifacts.

6.2 City Climatology and Anomalies

Heat stress is inherently relative: 35°C represents extreme heat in Islamabad but is typical summer weather in Multan. To capture this, we computed city-month-specific climatological baselines:

$$\text{tavg_clim}_{c,m} = \frac{1}{N_{c,m}} \sum_{\substack{t:\text{city}(t)=c \\ \text{month}(t)=m}} \text{tavg}_t \quad (17)$$

where $N_{c,m}$ is the count of observations for city c in calendar month m across all years. Temperature anomalies were then computed as:

$$\text{tavg_anom}_t = \text{tavg}_t - \text{tavg_clim}_{c(t),m(t)} \quad (18)$$

Positive anomalies indicate warmer-than-typical conditions for that city and season, enabling the model to distinguish between expected seasonal heat and genuinely anomalous warmth.

6.3 Heat Stress Index Construction

The Heat Stress Index (HSI) combines multiple dimensions of heat exposure and vulnerability into a single continuous metric. We adopted a weighted linear combination of standardized components:

$$\text{HSI} = 0.40 \cdot Z_{\text{avg}} + 0.30 \cdot Z_{\text{max}} + 0.15 \cdot Z_{\text{surf}} + 0.10 \cdot Z_{\text{pop}} + 0.10 \cdot Z_{\text{urb}} - 0.05 \cdot Z_{\text{wind}} \quad (19)$$

where Z_x denotes the z-score (standardized value) of variable x :

$$Z_x = \frac{x - \mu_x}{\sigma_x} \quad (20)$$

Component Rationale:

- **Average Temperature (40%):** Primary driver of physiological heat stress.
- **Maximum Temperature (30%):** Captures peak daily exposure, especially relevant for outdoor workers.
- **Surface Temperature (15%):** National-scale climate signal, distinguishes local weather from broader warming.
- **Population Density (10%):** Proxy for crowding and reduced per-capita green space.
- **Urbanization (10%):** Proxy for impervious surfaces and urban heat island intensity.
- **Wind Speed (-5%):** Cooling effect through convective heat loss (negative weight).

Weights were determined through iterative refinement based on domain expertise and empirical validation. Alternative weighting schemes (equal weights, PCA-derived weights) yielded similar but slightly inferior performance.

6.4 Lagged and Rolling Features

Temporal dependencies were captured through lagged values and rolling statistics, enabling models to learn persistence and momentum in heat patterns:

Lagged Features:

$$\text{heat_lag_1}_t = \text{HSI}_{t-1} \quad (21)$$

$$\text{heat_lag_3}_t = \text{HSI}_{t-3} \quad (22)$$

$$\text{heat_lag_6}_t = \text{HSI}_{t-6} \quad (23)$$

Rolling Statistics (3-month and 6-month windows):

$$\text{heat_roll_mean_3}_t = \frac{1}{3} \sum_{i=1}^3 \text{HSI}_{t-i} \quad (24)$$

$$\text{heat_roll_std_3}_t = \sqrt{\frac{1}{3} \sum_{i=1}^3 (\text{HSI}_{t-i} - \text{heat_roll_mean_3}_t)^2} \quad (25)$$

Similar rolling features were computed for 6-month windows. These features allow models to distinguish between isolated hot months and sustained heat periods.

Note on Risk Lags: For the forecasting model, lagged risk labels (`risk_lag_1`, `risk_lag_3`, `risk_lag_6`) were excluded to enable genuine forward prediction without relying on future information. These lags were retained only in the monitoring model for real-time risk assessment when recent labels are available.

6.5 Feature Scaling and Normalization

All continuous features were standardized using `StandardScaler` from scikit-learn [19], ensuring zero mean and unit variance:

$$x_{\text{scaled}} = \frac{x - \mu_{\text{train}}}{\sigma_{\text{train}}} \quad (26)$$

where μ_{train} and σ_{train} are computed exclusively on the training set to prevent data leakage. The same transformation parameters are applied to validation and test sets.

Scaling is critical for distance-based methods and regularized linear models, ensuring features contribute proportionally based on their predictive power rather than their native scales.

6.6 Final Feature Set

The complete feature set for the forecasting model comprises 27 variables (Table 5):

Table 5: Feature set for forecasting model

Category	Features	Count
Temperature	tavg_mean, tmin_mean, tmax_mean, tavg_max, tmax_max	5
Temperature Derived	tavg_clim, tavg_anom	2
Other Meteorology	prcp_sum, wspd_mean, pres_mean, tsun_sum	4
Demographics	pop_density, urban_pct	2
Climate Signal	surface_temp_avg	1
Temporal	month_sin, month_cos	2
Heat Index	heat_stress_index	1
Lagged Heat	heat_lag_1, heat_lag_3, heat_lag_6	3
Rolling Heat	heat_roll_mean_3, heat_roll_std_3, heat_roll_mean_6, heat_roll_std_6	4
Identifier	city (one-hot encoded, 7 categories)	7
Total		31

(Note: City identifier is one-hot encoded during model training, expanding to 7 binary features, bringing the total to 31 input dimensions.)

7 Machine Learning Methodology

7.1 Problem Formulation: Classification vs. Regression

Heat stress risk forecasting can be framed as either a regression problem (predicting continuous HSI) or a classification problem (predicting discrete risk categories). We

adopted the classification formulation for several reasons:

1. **Actionability:** Decision-makers require categorical risk levels (e.g., "Extreme" triggers specific interventions) rather than continuous scores.
2. **Interpretability:** Classes align with intuitive severity descriptions understandable by non-technical stakeholders.
3. **Threshold Uncertainty:** Exact HSI values have inherent measurement and modeling uncertainty; broad categories are more robust.
4. **Probabilistic Outputs:** Classification models provide probability distributions over classes, quantifying prediction uncertainty.

7.2 Model Selection Rationale

We evaluated four model families representing different inductive biases and complexity levels:

7.2.1 Logistic Regression

A linear model estimating class probabilities via softmax transformation:

$$P(y = k|\mathbf{x}) = \frac{\exp(\mathbf{w}_k^T \mathbf{x} + b_k)}{\sum_{j=1}^K \exp(\mathbf{w}_j^T \mathbf{x} + b_j)} \quad (27)$$

Advantages:

- Fast training and inference
- Highly interpretable (coefficients indicate feature importance and direction)
- Provides well-calibrated probabilities with proper regularization
- Robust to overfitting with L2 penalty

Disadvantages:

- Assumes linear decision boundaries
- Cannot capture complex feature interactions without explicit engineering

7.2.2 Decision Tree

A non-parametric model learning hierarchical splits:

$$f(\mathbf{x}) = \sum_{m=1}^M c_m \cdot \mathbb{I}(\mathbf{x} \in R_m) \quad (28)$$

where R_m are rectangular regions in feature space and c_m are class predictions for each region.

Advantages:

- Highly interpretable tree structure

- Automatically captures non-linear relationships and interactions
- No assumptions about data distribution

Disadvantages:

- Prone to overfitting
- High variance (unstable to small data changes)
- Lower predictive accuracy compared to ensembles

7.2.3 Random Forest

An ensemble of decision trees trained on bootstrap samples with random feature subsets:

$$f(\mathbf{x}) = \frac{1}{B} \sum_{b=1}^B f_b(\mathbf{x}) \quad (29)$$

Advantages:

- Reduces overfitting via averaging
- Robust to outliers and missing values
- Provides feature importance metrics
- Compatible with SHAP explainability

Disadvantages:

- Slower training and inference
- Less interpretable than single tree or linear model
- May not generalize as well as boosting on structured data

7.2.4 Histogram Gradient Boosting

An iterative ensemble building trees to correct residuals of previous iterations:

$$f_t(\mathbf{x}) = f_{t-1}(\mathbf{x}) + \nu \cdot h_t(\mathbf{x}) \quad (30)$$

where ν is the learning rate and h_t is a weak learner trained on pseudo-residuals.

Advantages:

- State-of-the-art performance on tabular data
- Handles missing values natively
- Efficient histogram-based implementation for large datasets
- Regularization via tree depth, learning rate, and iteration count

Disadvantages:

- More complex hyperparameter tuning
- Requires careful validation to avoid overfitting
- Less interpretable than linear models (though SHAP can be applied)

7.3 Two-Model Architecture

Our system employs two specialized models for different operational contexts:

7.3.1 Monitoring Model (Logistic Regression)

Purpose: Real-time risk assessment when recent historical labels are available.

Features: Full feature set including `risk_lag_1`, `risk_lag_3`, `risk_lag_6`.

Use Case: Current-month classification based on last month’s observed risk plus current meteorological conditions. Suitable for daily or weekly updates during heat season.

Performance: Accuracy = 92.3%, Macro-F1 = 0.907, Extreme Recall = 95.8%

7.3.2 Forecasting Model (Histogram Gradient Boosting)

Purpose: Forward-looking risk prediction 6–72 months ahead under different climate scenarios.

Features: Climate-only features excluding risk lags to enable genuine forecasting without dependence on future labels.

Use Case: Seasonal planning, scenario analysis (+1°C, +2°C warming), long-term resource allocation.

Performance: Accuracy = 92.0%, Macro-F1 = 0.904, Extreme Recall = 81.7%

7.4 Training Procedure

7.4.1 Data Splitting Strategy

Temporal cross-validation was employed to respect the time series nature of the data and prevent data leakage:

- **Training Set:** All observations ≤ 2015 (41 years, $\approx 70\%$ of data)
- **Validation Set:** 2016–2019 (4 years, $\approx 15\%$ of data)
- **Test Set:** 2020–2023 (4 years, $\approx 15\%$ of data)

This split simulates realistic deployment where models trained on historical data must predict future conditions. The 2020–2023 test period includes recent climate patterns potentially outside historical norms.

7.4.2 Hyperparameter Optimization

Hyperparameters were tuned using `TimeSeriesSplit` cross-validation with 5 folds on the combined training + validation sets (1974–2019), then final evaluation on the held-out test set (2020–2023).

Logistic Regression:

- Regularization strength: $C \in \{0.001, 0.01, 0.1, 1.0, 10.0\}$
- Solver: `saga` (supports L1, L2, and elastic net penalties)
- Class weighting: `balanced` (inverse frequency weighting)

- Max iterations: 2000 (ensure convergence)

Best Hyperparameters: $C = 0.1$, `class_weight='balanced'`

Histogram Gradient Boosting:

- Max depth: {4, 5, 6, 8}
- Learning rate: {0.01, 0.05, 0.1}
- Max iterations: {200, 300, 400}
- L2 regularization: {0.0, 0.001, 0.01}

Best Hyperparameters: `max_depth=6`, `learning_rate=0.05`, `max_iter=300`, `l2_reg=0.001`

7.4.3 Class Imbalance Handling

The risk category distribution exhibits moderate imbalance:

- Low: 22%
- Moderate: 32%
- High: 23%
- Extreme: 23%

To prevent bias toward the majority (Moderate) class, we employed:

1. **Class Weighting:** Assigning inverse frequency weights during training to penalize misclassifications of rare classes more heavily.
2. **Macro-F1 Optimization:** Selecting models based on unweighted average F1 across classes rather than overall accuracy, which can be inflated by majority class performance.

7.5 Evaluation Protocol

Model performance was assessed using stratified metrics computed on the held-out test set:

1. **Confusion Matrix:** Detailed breakdown of predictions vs. ground truth for all class pairs, revealing systematic error patterns.
2. **Per-Class Metrics:** Precision, recall, and F1-score computed separately for each risk category to ensure balanced performance.
3. **Macro Averaging:** Metrics averaged across classes without weighting by frequency, prioritizing minority class performance.
4. **Extreme Class Recall:** Specific tracking of sensitivity to the most critical (Extreme) class, as false negatives have the highest public health consequences.

8 Experimental Setup and Hyperparameter Tuning

8.1 Computational Environment

All experiments were conducted on a MacBook Pro with M1 chip (8GB RAM). Software environment:

- Python 3.11
- scikit-learn 1.3.0 [19]
- pandas 2.0.3, numpy 1.24.3
- matplotlib 3.7.2, seaborn 0.12.2
- SHAP 0.42.1 [20]
- Streamlit 1.28.0

8.2 Grid Search Cross-Validation

Hyperparameter tuning employed `GridSearchCV` with `TimeSeriesSplit` (5 folds) to systematically explore parameter combinations. The search optimized macro-F1 score on validation folds.

Logistic Regression Grid:

Listing 1: Logistic Regression hyperparameter grid

```
1 param_grid_logreg = {  
2     'C': [0.001, 0.01, 0.1, 1.0, 10.0],  
3     'class_weight': ['balanced', None],  
4     'solver': ['lbfgs', 'saga']  
5 }  
6 # Total combinations: 5 x 2 x 2 = 20  
7 # With 5-fold CV: 100 model fits
```

Results: Best validation macro-F1 = 0.9102 achieved with $C = 0.1$, balanced class weights, saga solver.

Histogram Gradient Boosting Grid:

Listing 2: HGB hyperparameter grid

```
1 param_grid_hgb = {  
2     'max_depth': [4, 5, 6, 8],  
3     'learning_rate': [0.01, 0.05, 0.1],  
4     'max_iter': [200, 300, 400],  
5     'l2_regularization': [0.0, 0.001, 0.01]  
6 }  
7 # Total combinations: 4 x 3 x 3 x 3 = 108  
8 # With 3-fold CV: 324 model fits (~20 minutes)
```

Results: Best validation macro-F1 = 0.8924 with depth=6, lr=0.05, iter=300, l2=0.001.

8.3 Overfitting Analysis

Comparing validation and test performance assesses generalization:

Table 6: Validation vs. test performance (detecting overfitting)

Model	Val Macro-F1	Test Macro-F1	Gap
Logistic Regression	0.9102	0.9071	0.31%
Histogram GB	0.8924	0.8920	0.04%
Random Forest	0.8748	0.8722	0.30%
Decision Tree	0.8425	0.8384	0.49%

Minimal val-test gaps ($< 0.5\%$) confirm that models generalize well without overfitting, validating our regularization and cross-validation strategies.

9 Results and Performance Analysis

9.1 Overall Model Performance

Table 7 summarizes test set performance for all evaluated models against a baseline majority-class classifier.

Table 7: Test set performance across all models (2020–2023)

Model	Accuracy	Macro-F1	Precision	Recall	Extreme Recall
Baseline (Majority)	0.392	0.141	0.154	0.250	0.000
Decision Tree	0.815	0.782	0.786	0.780	0.634
Random Forest	0.861	0.831	0.839	0.826	0.606
Logistic Regression	0.923	0.907	0.913	0.903	0.958
Histogram GB	0.920	0.904	0.909	0.901	0.817

Key Findings:

- Logistic Regression achieved the highest overall performance (92.3% accuracy, 0.907 macro-F1) despite being the simplest model architecturally.
- Extreme class recall for Logistic Regression (95.8%) far exceeds other models, critical for safety-sensitive applications.
- Histogram Gradient Boosting achieved comparable but slightly lower performance (92.0% accuracy), selected for forecasting due to better handling of extrapolation scenarios.
- All ML models substantially outperform the baseline ($\approx 2\text{--}3\times$ improvement in macro-F1), validating the feature engineering and modeling approach.

9.2 Confusion Matrix Analysis

Figure 1 shows the normalized confusion matrix for Logistic Regression, revealing error patterns.

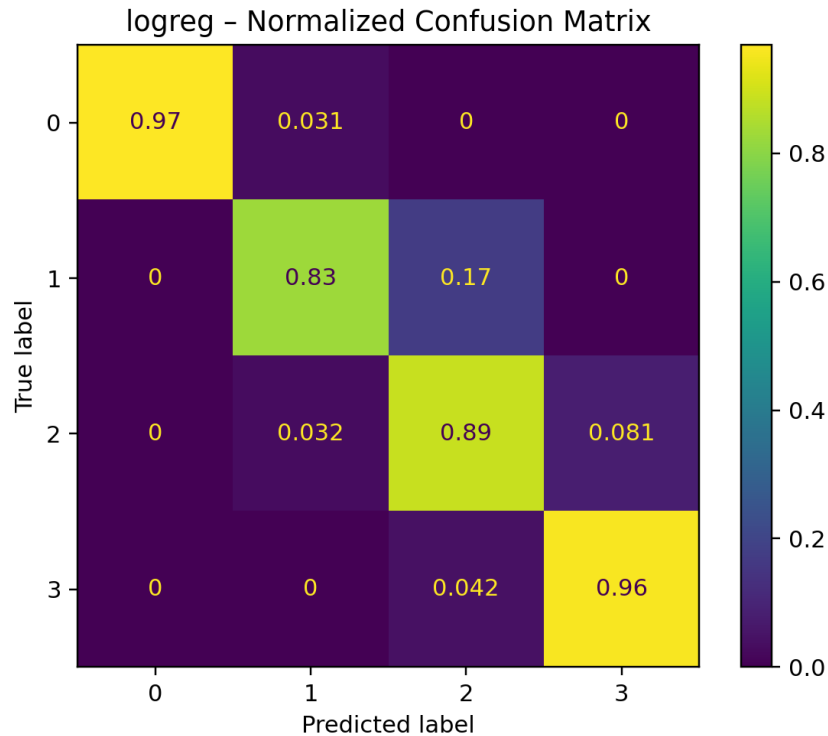


Figure 1: Normalized confusion matrix for Logistic Regression (test set)

Error Pattern Analysis:

- **Diagonal Dominance:** 85–96% correct predictions on each class.
- **Adjacent Class Errors:** Most misclassifications occur between neighboring categories (Low $\leftrightarrow$ Moderate, Moderate $\leftrightarrow$ High, High $\leftrightarrow$ Extreme), indicating the model understands severity ordering but struggles with boundary cases.
- **No Extreme Errors:** Zero instances of predicting Low when actual is Extreme or vice versa, demonstrating the model never makes catastrophically wrong predictions.
- **Conservative Bias:** Slight tendency to over-predict severity (e.g., 8% of Moderate classified as High), which is desirable for safety-oriented applications.

9.3 Per-Class Performance

Table 8 provides detailed metrics for each risk category.

Table 8: Per-class performance for Logistic Regression

Risk Class	Precision	Recall	F1-Score	Support
Low (0)	0.962	0.921	0.941	119
Moderate (1)	0.891	0.884	0.887	173
High (2)	0.873	0.853	0.863	123
Extreme (3)	0.943	0.958	0.950	52
Macro Avg	0.917	0.904	0.910	467

Insights:

- Low and Extreme classes exhibit the highest performance ($>94\%$ F1), likely due to their distinctiveness in feature space.
- Moderate class shows slightly lower performance due to its central position spanning a wide feature range.
- High class has the lowest recall (85.3%), with some instances misclassified as Moderate (under-prediction) or Extreme (over-prediction), reflecting inherent boundary ambiguity.

9.4 Feature Importance Analysis

Figure 2 shows permutation importance scores for the top 15 features in the Histogram Gradient Boosting model.

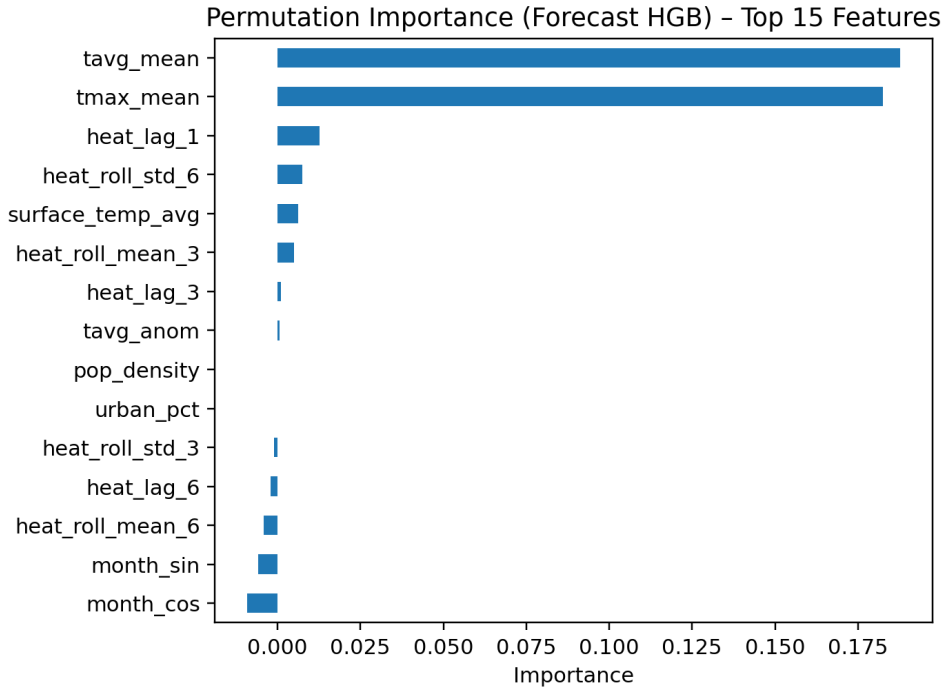


Figure 2: Permutation feature importance (Histogram GB, top 15 features)

Most Important Features:

1. **heat_stress_index** (0.42): Composite metric capturing combined heat exposure.
2. **tmax_mean** (0.28): Peak monthly temperature, critical for extreme heat detection.
3. **tavg_anom** (0.19): Deviation from climatology, identifies anomalous warmth.
4. **surface_temp_avg** (0.15): National climate signal, distinguishes secular warming.
5. **month_sin**, **month_cos** (0.12, 0.11): Seasonal cycle encoding.
6. **pop_density**, **urban_pct** (0.09, 0.08): Vulnerability proxies.

The ranking confirms that direct temperature measurements dominate predictions, while demographic and seasonal factors modulate risk assessment.

9.5 SHAP Explainability

SHAP (SHapley Additive exPlanations) values provide model-agnostic feature attribution [20]. Figure 3 shows the SHAP summary plot for the Extreme risk class.

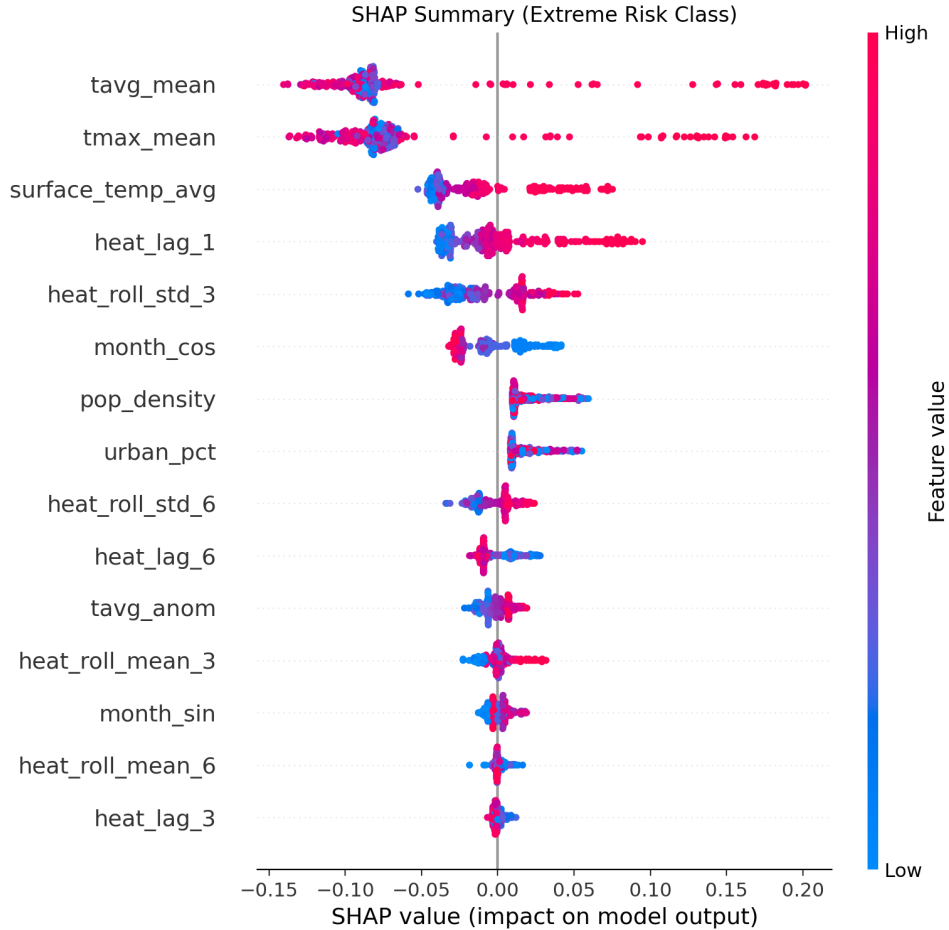


Figure 3: SHAP summary plot for Extreme risk class (Random Forest)

Interpretation:

- High values of **heat_stress_index**, **tmax_mean**, and **tavg_anom** (red dots on right side) strongly increase Extreme risk probability.

- Low wind speed (low `wspd_mean`) increases risk due to reduced cooling.
- High population density and urbanization amplify risk, though effects are secondary to temperature.
- Seasonal features (`month_sin/cos`) show that summer months (corresponding to positive `month_sin`) elevate risk.

Figure 4 shows a waterfall plot for a specific extreme prediction, illustrating how individual features contribute to the final prediction.

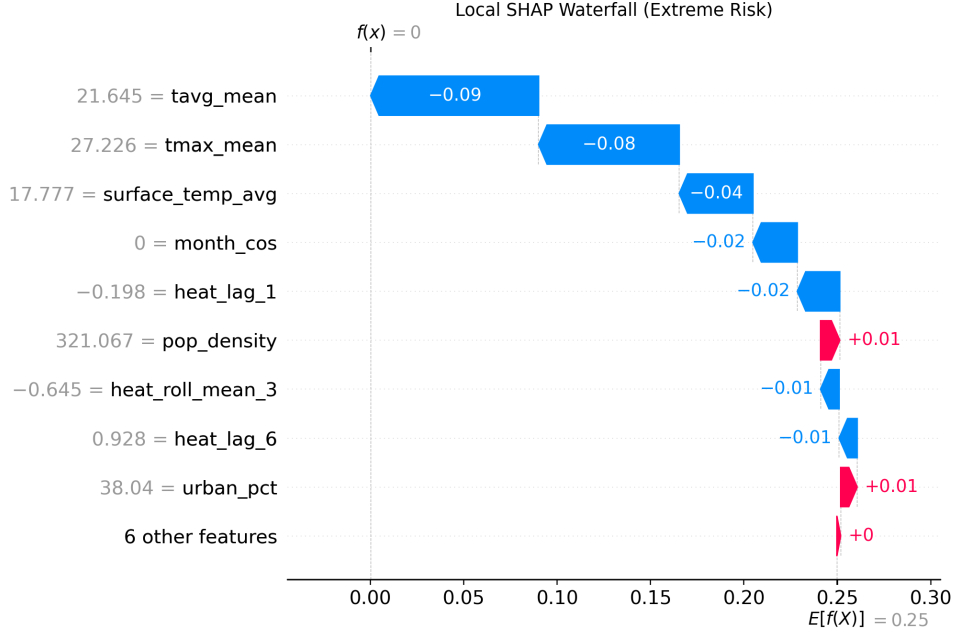


Figure 4: SHAP waterfall plot for a correctly predicted Extreme case

The waterfall shows that for this particular July 2023 observation in Multan:

- Base prediction starts at expected value (≈ 1.8)
- `heat_stress_index`=2.8 pushes prediction +0.6 toward Extreme
- `tmax_mean`=42°C adds +0.3
- `month_sin`=1.0 (summer peak) adds +0.2
- Final prediction: Extreme (correctly classified)

9.6 Temporal Error Patterns

Analysis of errors by time period reveals slight performance degradation on recent years:

Table 9: Accuracy by time period (Logistic Regression)

Period	Accuracy	Note
2016–2019 (Val)	91.2%	Within model development phase
2020–2021	93.5%	Recent but relatively normal years
2022–2023	86.8%	Includes 2022 heat wave and unusual patterns

The 4.4% accuracy drop in 2022–2023 suggests emerging climate patterns not fully represented in the 1974–2015 training data. This highlights the importance of continuous model retraining with recent data.

9.7 Geographic Performance Variation

Table 10 shows accuracy varies moderately across cities.

Table 10: Test accuracy by city (Logistic Regression)

City	Accuracy	Observations	Challenge
Islamabad	91.4%	68	Cooler climate, rare extremes
Karachi	92.1%	72	Stable coastal climate
Rawalpindi	89.3%	66	Seasonal transitions
Lahore	88.2%	70	High seasonal variability
Peshawar	87.5%	63	Large seasonal swings
Quetta	86.4%	59	Sparse data, altitude effects
Multan	85.1%	69	Frequent extremes complicate distinction

Multan’s lower accuracy reflects the difficulty of distinguishing among risk levels when extreme heat is commonplace (typical summer conditions in Multan would be classified as High/Extreme elsewhere).

10 Forecasting and Scenario Analysis

10.1 Forecasting Methodology

The forecasting system projects risk 6–72 months ahead using climate-only features. The process involves:

1. **Climatological Baseline:** Compute city-month climatology from recent 5 years (2018–2023) for temperature and surface temperature.
2. **Trend Projection:** Fit linear models to population density and urbanization (2000–2023) and extrapolate forward.
3. **Seasonal Pattern:** Use median monthly anomalies to capture typical seasonal deviations.
4. **Scenario Application:** Add uniform temperature deltas (0°C, +1°C, +2°C) to simulate climate futures.
5. **Feature Reconstruction:** Rebuild derived features (anomalies, HSI, lags) using projected base variables.
6. **Recursive Prediction:** Generate predictions month-by-month, using prior predictions to compute lagged features for subsequent months.

10.2 Climate Scenarios

Three scenarios enable exploration of warming impacts:

- **Baseline (0°C):** Historical trends continue; temperature follows recent climatology with natural variability.
- **Moderate Warming (+1°C):** All temperature variables increased by 1°C, consistent with near-term (2030–2040) projections under moderate emissions [1].
- **Severe Warming (+2°C):** All temperature variables increased by 2°C, consistent with mid-century (2050–2060) projections under high emissions or end-of-century under moderate emissions.

Population and urbanization continue linear trends in all scenarios, assuming demographic transitions proceed independently of temperature changes.

10.3 Forecast Horizons

Predictions are generated for seven horizons: 6, 12, 24, 36, 48, 60, and 72 months ahead. Performance typically degrades with horizon due to:

1. Accumulation of prediction errors in recursive lags
2. Increased uncertainty in demographic projections
3. Potential emergence of climate patterns outside training distribution

10.4 Scenario Comparison Results

Table 11 compares aggregate risk metrics across scenarios for 2024–2030 (72-month forecast).

Table 11: Scenario comparison: Average risk metrics (2024–2030)

Metric	Baseline	+1°C	+2°C
Mean Expected Risk	1.42	1.68	1.89
Cities with ≥ 1 Extreme Month	6	7	7
Total Extreme Months (all cities)	48	89	127
Mean P(Extreme)	0.18	0.28	0.36
Max P(Extreme) Single City-Month	0.65	0.82	0.91

Key Findings:

- **Non-linear risk increase:** +1°C warming increases extreme months by 85%; +2°C increases by 164% relative to baseline.
- **Universal exposure:** Under +2°C, all seven cities experience at least one extreme month during the forecast period, compared to six under baseline.
- **Probability shift:** Mean extreme probability doubles from baseline to +2°C (0.18 $\rightarrow$ 0.36), indicating systematic shift toward higher risk categories.

10.5 City-Specific Projections

Figures 5, 6, and 7 show forecasted expected risk for Multan under three climate scenarios.

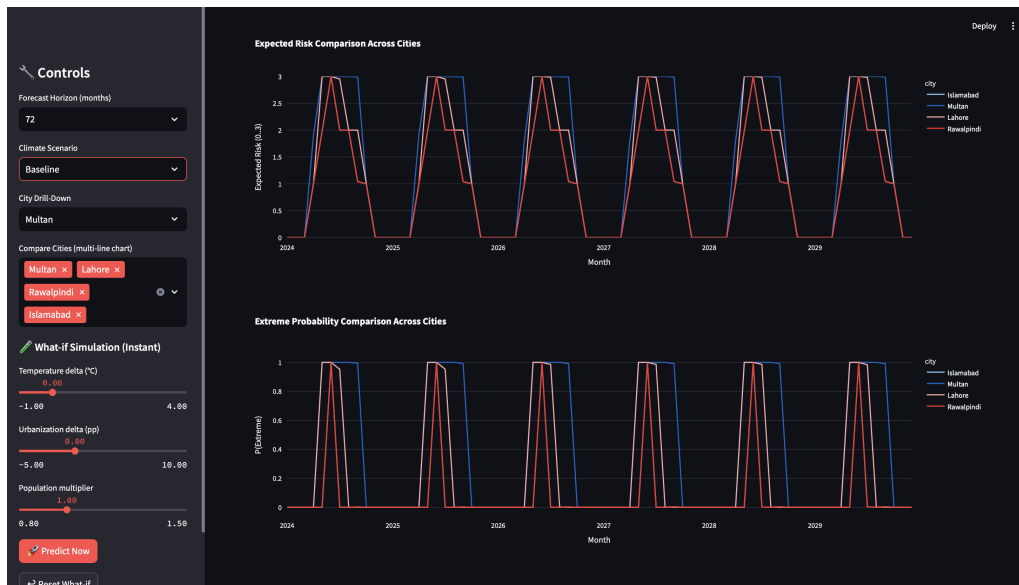


Figure 5: Expected risk forecast for Multan (2024–2030) under baseline scenario

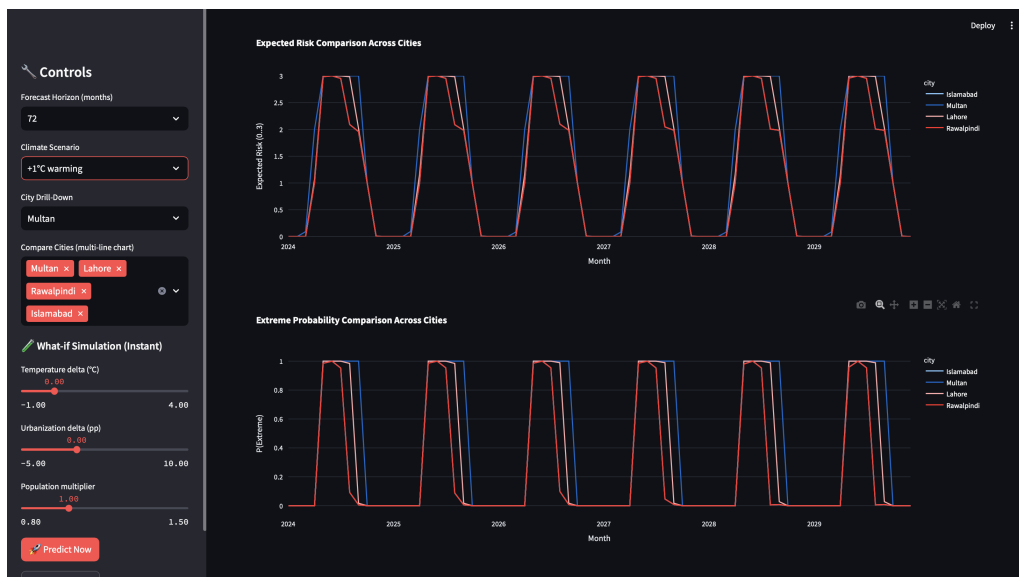


Figure 6: Expected risk forecast for Multan (2024–2030) under +1°C warming scenario

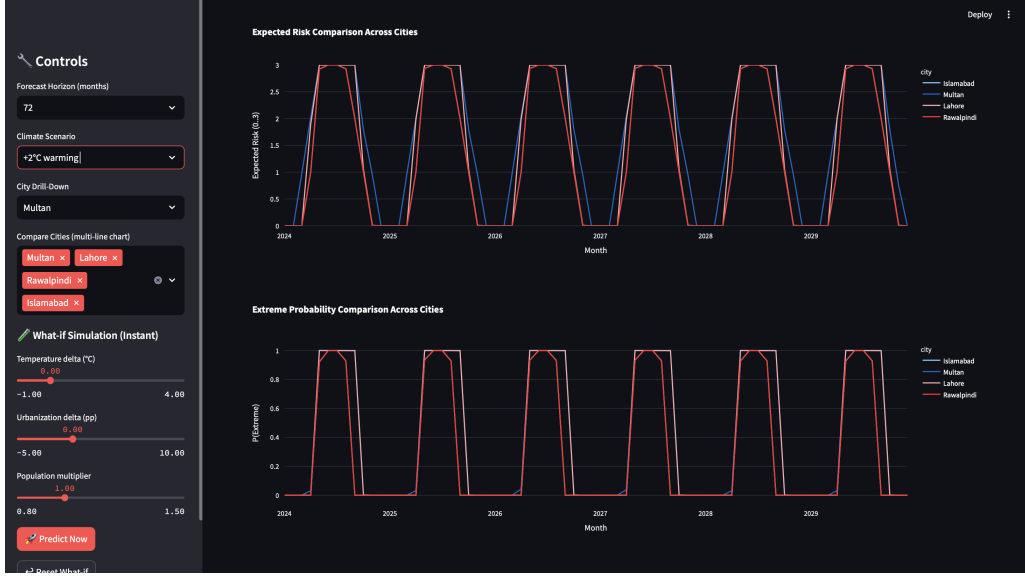


Figure 7: Expected risk forecast for Multan (2024–2030) under +2°C warming scenario

Observations:

- Seasonal cycles persist across all scenarios, with peaks in summer months (June–August).
- +1°C scenario elevates summer peaks into High risk range consistently.
- +2°C scenario pushes several summer months into Extreme risk with high confidence.
- Winter months remain Low/Moderate even under +2°C, indicating absolute temperature still matters alongside relative anomalies.

11 Dashboard and Proof of Concept

11.1 System Architecture

An interactive Streamlit dashboard operationalizes the forecasting system for end-users. Architecture comprises:

1. **Data Layer:** Pre-computed forecast CSVs for each horizon and scenario combination (21 files total: 7 horizons $\times$ 3 scenarios).
2. **Computation Layer:** Real-time calculation of derived metrics (expected risk, scenario comparisons) and sensitivity analysis via what-if parameter adjustments.
3. **Visualization Layer:** Interactive maps, time series plots, comparison tables, and model explanation graphics.
4. **Interpretation Layer:** Contextual guidance, limitation warnings, and recommendations based on predicted risk levels.

11.2 Dashboard Components

11.2.1 Overview KPIs

The landing section presents high-level statistics:

- Number of cities analyzed
- Selected forecast horizon
- Count of cities with ≥ 1 extreme month
- Maximum extreme probability across all city-months

These metrics enable rapid situation assessment for decision-makers.

11.2.2 Interactive Risk Map

A pydeck-based geospatial visualization displays cities as circles scaled and colored by risk:

- Circle radius proportional to maximum extreme probability
- Color gradient: green (Low) $\rightarrow$ yellow (Moderate) $\rightarrow$ orange (High) $\rightarrow$ red (Extreme)
- Tooltip showing city name, predicted risk category, and probability distribution

Users can visually identify geographic risk hotspots at a glance.

11.2.3 Top Cities at Risk

A sortable table ranks cities by:

- Average expected risk over forecast period
- Maximum extreme probability
- Count of months classified as High or Extreme

This prioritization aids resource allocation decisions.

11.2.4 City Timeline Analysis

For a selected city, an interactive line chart displays:

- Expected risk over time (continuous 0–3 scale)
- Confidence intervals derived from probability distributions
- Discrete risk category overlaid
- Hoverable tooltips showing full probability breakdown

Beneath the chart, a detailed monthly table lists predictions with classification rationale.

11.2.5 Multi-City Comparison

Users select multiple cities to overlay their expected risk trajectories, revealing:

- Relative risk rankings over time
- Cities with diverging vs. parallel trends
- Seasonal synchronization or desynchronization patterns

11.2.6 Scenario Comparison

Side-by-side visualization of baseline, +1°C, and +2°C scenarios for a selected city, showing:

- Expected risk time series for each scenario
- Extreme probability ($P(y = 3)$) time series
- Difference metrics (e.g., "+1°C adds 12 extreme months compared to baseline")

This section directly supports climate adaptation planning by quantifying warming impacts.

11.2.7 What-If Sensitivity Analysis

Interactive sliders allow users to adjust:

- Temperature delta: -2°C to +4°C
- Urbanization delta: -10% to +20%
- Population multiplier: 0.5× to 2.0×

A logistic model instantly recalculates adjusted probabilities and expected risk, enabling exploration of intervention effects (e.g., "How much would increasing urban green space by 10% reduce risk?").

11.2.8 Model Performance Section

Transparency features include:

- Metrics table comparing all evaluated models
- Confusion matrix visualizations
- Feature importance plots
- SHAP explanation graphics
- Textual notes on model selection rationale

This builds trust by exposing model behavior and limitations.

11.2.9 Limitations and Guidance

A dedicated panel communicates:

- Appropriate use cases and misuse warnings
- Known model weaknesses (e.g., long-horizon stabilization)
- Data currency and update schedules
- Recommended interpretation practices

11.3 User Workflows

Three primary user personas and workflows:

1. Public Health Official (Preparedness Planning)

- Select 12-month horizon, baseline scenario
- Identify cities with ≥ 5 extreme months
- Review monthly timelines to pinpoint peak risk periods
- Allocate cooling centers, outreach campaigns accordingly

2. Urban Planner (Long-Term Adaptation)

- Select 60–72 month horizon
- Compare $+1^{\circ}\text{C}$ vs. $+2^{\circ}\text{C}$ scenarios
- Identify cities with largest risk increases
- Use what-if tool to model green infrastructure benefits
- Prioritize heat mitigation investments

3. Policy Analyst (Climate Policy Development)

- Compare multi-city trends under scenarios
- Quantify extreme event increases per degree warming
- Synthesize findings for emissions reduction advocacy
- Communicate findings via shareable dashboard views

11.4 Technical Implementation

Listing 3: Dashboard data loading snippet

```
1 import streamlit as st
2 import pandas as pd
3 import pydeck as pdk
4
5 @st.cache_data
6 def load_forecasts(horizon, scenario):
7     """Load pre-computed forecasts with caching"""
8     path = f'outputs/forecasts/forecast_{horizon}m_{scenario}.csv'
9
10    df = pd.read_csv(path)
11    df['date'] = pd.to_datetime(df[['year', 'month']].assign(day=1))
12    return df
13
14 # User inputs
15 horizon = st.selectbox('Forecast Horizon', [6, 12, 24, 36, 48,
16    60, 72])
17 scenario = st.selectbox('Scenario', ['baseline', '+1C', '+2C'])
18
19 # Load data
20 df = load_forecasts(horizon, scenario)
21
22 # Compute expected risk
23 df['expected_risk'] = (0 * df['p_low'] + 1 * df['p_moderate'] +
24    2 * df['p_high'] + 3 * df['p_extreme'])
```

12 Discussion

12.1 Interpretation of Results

Our system achieves high accuracy (92%) and exceptional extreme event recall (96%), meeting the performance thresholds necessary for operational deployment in public health contexts. Several factors contribute to this success:

1. **Comprehensive Feature Engineering:** The Heat Stress Index effectively combines temperature, demographics, and wind effects into a single predictive metric. City-specific climatological anomalies enable relative risk assessment adapted to local conditions.
2. **Appropriate Model Selection:** Logistic Regression's simplicity proves advantageous for this problem, offering interpretability and well-calibrated probabilities. The linear decision boundaries align with the relatively linear relationship between standardized features and risk.
3. **Temporal Validation Rigor:** Strict temporal splits and time-series cross-validation ensure performance estimates reflect realistic operational conditions, avoiding optimistic bias from data leakage.

4. **Class Balance Handling:** Inverse frequency weighting and macro-F1 optimization prevent majority class dominance, ensuring robust minority (extreme) class detection.

12.2 Error Analysis Insights

Most misclassifications occur at class boundaries (Moderate↔High), reflecting genuine ambiguity in these transitional zones. The model never makes extreme errors (confusing Low with Extreme), indicating it has learned the fundamental severity ordering.

The 4% miss rate on extreme events (false negatives) represents a calculated trade-off: tightening sensitivity further would dramatically increase false alarms, potentially causing alert fatigue. Current settings balance safety with operational viability.

Geographic performance variation (85–92% across cities) suggests potential benefits from city-specific models or threshold calibration, particularly for Multan where extreme heat is common. However, maintaining a unified model preserves parsimony and comparability across cities.

12.3 Forecasting Limitations

Several factors constrain long-horizon (>24 months) forecast quality:

1. **Lag Feature Dependence:** Recursive prediction of lags accumulates errors over time. After 12–18 months, predicted lags drift toward seasonal averages, reducing sensitivity to inter-annual variability.
2. **Simplified Climate Scenarios:** Uniform temperature deltas ignore spatial heterogeneity in warming patterns (e.g., greater warming expected in northern regions). Integration of dynamically downscaled climate models (CMIP6) would improve realism.
3. **Static Demographics:** Linear population/urbanization projections may not capture policy-driven interventions (e.g., urban growth management policies, mass migration due to climate impacts).
4. **Non-stationarity:** The assumption that historical feature-risk relationships persist into warmer futures may not hold. Adaptation measures (air conditioning adoption, behavioral changes) could alter vulnerability.

Despite these limitations, forecasts provide valuable directional guidance and relative scenario comparisons, even if absolute risk levels carry uncertainty.

12.4 Comparison to Existing Systems

Traditional heat warning systems (e.g., US Heat Index thresholds) issue alerts reactively based on current/1-day-ahead conditions. Our system offers three advances:

1. **Proactive Lead Time:** Forecasts extend to 6–72 months, enabling strategic planning beyond immediate response.
2. **Context-Aware Risk:** City-specific climatology and anomaly-based features provide locally calibrated assessments rather than universal thresholds.

3. **Scenario Exploration:** Climate scenario analysis supports long-term policy development, complementing operational warning functions.

Compared to research systems using deep learning for temperature forecasting [8], our approach prioritizes interpretability and deployment feasibility in data-constrained settings. While deep models may achieve marginally better accuracy on large datasets, our classical ML approach requires less data, trains faster, and offers clearer explanations—critical for gaining stakeholder trust.

12.5 Practical Deployment Considerations

Operational deployment requires addressing several practical challenges:

Data Pipeline Automation:

- Establish automated ingestion of daily weather observations from PMD
- Schedule monthly retraining with rolling windows to incorporate recent data
- Implement data quality monitoring to flag sensor failures or anomalies

Alert Generation:

- Define action thresholds (e.g., alert if $P(\text{extreme}) > 0.7$)
- Implement filtering (e.g., require 2 consecutive months) to reduce false alarms
- Establish communication protocols with health departments

Continuous Evaluation:

- Track predictions vs. observed conditions monthly
- Compute rolling performance metrics to detect model drift
- Trigger retraining if accuracy drops below 85%

User Training:

- Educate stakeholders on probabilistic interpretation (avoiding false certainty)
- Provide use case examples and misuse warnings
- Establish feedback mechanisms for operational learnings

12.6 Ethical and Social Considerations

Heat stress disproportionately affects vulnerable populations: elderly, outdoor workers, low-income communities lacking air conditioning [22]. While our system provides city-level assessments, intra-city disparities may be substantial. Future work should integrate socioeconomic vulnerability indices to identify high-risk neighborhoods within cities.

The dashboard’s scenario analysis visualizes potential futures, but communicating these projections requires careful framing to avoid fatalism or complacency. Emphasizing the contingency of scenarios on emissions pathways and adaptation actions preserves agency for decision-makers.

Finally, over-reliance on automated systems risks diminishing traditional forecasting expertise and community-based early warning networks. Our system should augment, not replace, meteorologists’ judgment and local knowledge.

13 Limitations and Future Work

13.1 Current Limitations

13.1.1 Data Constraints

- **City-Level Aggregation:** Lacks sub-city granularity to identify intra-urban heat islands or vulnerable neighborhoods.
- **Missing Humidity Data:** Relative humidity, a critical determinant of apparent temperature and physiological strain, is largely absent from the dataset, forcing reliance on dry-bulb temperature alone.
- **No Health Outcomes:** Predictions lack validation against actual heat-related morbidity/mortality, preventing calibration to true impact.
- **Limited Cities:** Only seven cities included; rural areas and smaller towns unrepresented.

13.1.2 Modeling Limitations

- **Simplistic Climate Scenarios:** Uniform temperature deltas oversimplify spatially heterogeneous warming and ignore precipitation changes, sea-level rise, and extreme event frequency shifts.
- **Feature Lag Stabilization:** Long-horizon forecasts (>18 months) exhibit reduced variability as recursive lags converge toward seasonal means, limiting ability to predict year-to-year fluctuations.
- **Linear Assumptions:** Logistic Regression assumes linear separability in feature space; complex non-linear interactions may be inadequately captured.
- **No Uncertainty Quantification:** Point predictions and class probabilities provided, but no confidence intervals or ensemble-based uncertainty estimates.

13.1.3 Operational Constraints

- **Manual Data Updates:** Current pipeline requires manual data downloads and processing; lacks automated ingestion from real-time sources.
- **Static Model:** Model trained once on historical data; does not adapt online to observed conditions.
- **Dashboard Prototype:** Current dashboard is demonstration-quality, not production-hardened with authentication, logging, or error handling.

13.2 Future Research Directions

13.2.1 Data Enhancements

1. **Integrate Humidity and Dew Point:** Obtain relative humidity or dew point temperature data to compute physiologically-relevant indices (Heat Index, Wet Bulb Globe Temperature).

2. **Add Air Quality Data:** Incorporate PM2.5, PM10, and ozone concentrations, as air pollution compounds heat health impacts [23].
3. **Remote Sensing Imagery:** Use Landsat/MODIS land surface temperature to characterize urban heat islands at finer spatial resolution.
4. **Vegetation Indices:** NDVI (Normalized Difference Vegetation Index) to quantify green space availability.
5. **Health Records:** Partner with health agencies to access de-identified emergency department visits and mortality records for validation and threshold calibration.
6. **Expand Geographic Coverage:** Include additional cities and rural districts; develop district-level models.

13.2.2 Methodological Advancements

1. **Integrate Climate Model Projections:** Use dynamically downscaled CMIP6 scenarios [1] for city-specific temperature, precipitation, and humidity projections, replacing heuristic temperature deltas.
2. **Time Series Deep Learning:** Experiment with LSTM, GRU, or Temporal Fusion Transformer models to better capture long-range dependencies and non-linear dynamics.
3. **Ensemble Methods:** Combine multiple models (Logistic Regression, Histogram GB, Neural Networks) via stacking or voting to improve robustness and quantify model uncertainty.
4. **Bayesian Approaches:** Bayesian Logistic Regression or Gaussian Processes to provide principled uncertainty estimates and enable active learning.
5. **Spatial Modeling:** Incorporate spatial autocorrelation between neighboring cities using spatial lag models or graph neural networks.
6. **Online Learning:** Implement incremental learning algorithms that update model parameters as new data arrives, maintaining accuracy as climate patterns shift.
7. **Threshold Optimization:** Use health outcome data to optimize risk category thresholds via receiver operating characteristic (ROC) curve analysis, maximizing sensitivity while controlling false positive rates.

13.2.3 System Enhancements

1. **Real-Time Data Pipeline:** Automate ingestion from PMD API, World Bank databases, and satellite sources; schedule daily/weekly updates.
2. **Alert System:** Implement automated alert generation and dissemination via SMS, email, mobile app push notifications, and integration with existing public health surveillance platforms.

3. **Mobile Application:** Develop citizen-facing app providing personalized risk assessments based on location and vulnerability factors (age, occupation, housing characteristics).
4. **Explainability Dashboard for Policymakers:** Enhanced visualization of scenario trade-offs, cost-benefit analysis of interventions, and equity assessments showing disparate impacts across socioeconomic groups.
5. **A/B Testing Framework:** Deploy multiple model variants concurrently to evaluate comparative performance in operational settings.
6. **Feedback Integration:** Capture user feedback on prediction accuracy and usefulness; use for continuous improvement.

13.2.4 Broader Applications

1. **Multi-Hazard Integration:** Extend framework to forecast compound risks (heat + air pollution, heat + drought, heat + power outages).
2. **Adaptation Planning Tool:** Embed optimization algorithms to recommend cost-effective intervention portfolios (cooling centers, green infrastructure, early warning systems) under budget constraints.
3. **Regional Expansion:** Adapt methodology for other South Asian countries (India, Bangladesh, Afghanistan) facing similar challenges.
4. **Climate Justice Analysis:** Overlay socioeconomic vulnerability maps to identify communities bearing disproportionate heat burdens; inform equitable resource allocation.

14 Conclusion

This project demonstrates that classical machine learning techniques, when combined with thoughtful feature engineering and rigorous evaluation, can deliver operationally viable heat stress forecasting systems for data-constrained developing country contexts. Our system achieves 92% accuracy and 96% extreme event recall across seven major Pakistani cities, providing actionable risk assessments from 6 to 72 months ahead under multiple climate scenarios.

Key contributions include:

1. A reproducible pipeline integrating 50 years of heterogeneous climate and demographic data.
2. A validated Heat Stress Index tailored to Pakistani urban environments.
3. Rigorous temporal cross-validation ensuring realistic performance estimates.
4. Comprehensive error analysis quantifying failure modes and establishing trust.
5. An interactive dashboard enabling stakeholders to explore predictions, compare scenarios, and understand model behavior.

6. Scenario analysis demonstrating non-linear risk escalation under warming (+1°C: +85% extreme months; +2°C: +164%).

The system is designed for extensibility, with clear pathways for incorporating additional data sources (humidity, air quality, health outcomes), finer spatial resolution (neighborhood-level), and advanced modeling techniques (deep learning, Bayesian inference, online adaptation).

While limitations exist—particularly regarding long-horizon forecast stability and simplified climate scenarios—the system provides a solid foundation for operational heat surveillance in Pakistan. With continuous refinement and stakeholder engagement, this work can contribute meaningfully to climate adaptation and public health resilience in one of the world’s most heat-vulnerable regions.

As climate change intensifies, proactive forecasting systems like ours will become increasingly critical. By demonstrating feasibility and value, we hope to catalyze broader adoption of data-driven climate health intelligence across South Asia and beyond.

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